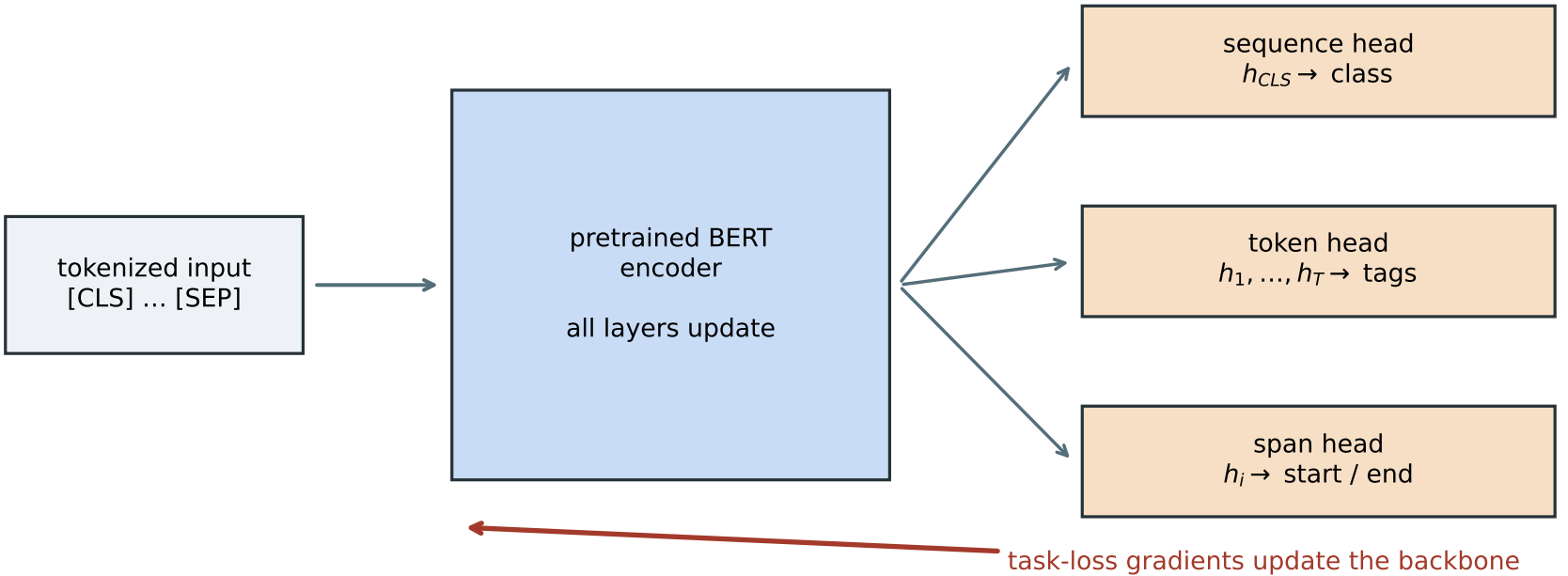
tokenized input
[CLS] ... [SEP]
pretrained BERT
encoder
all layers update
sequence head
$h_{CLS} \rightarrow$ class
token head
$h_1, \ldots, h_T \rightarrow$ tags
span head
$h_i \rightarrow$ start / end
task-loss gradients update the backbone